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Nonholonomic Path Planning Optimization for Dubins' Vehicles

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Abstract— **In this paper we deal with the problem of path length optimization for nonholonomic robots modeled as Dubins' vehicles. We present an improved solution that computes paths through two-dimensional waypoints, that are shorter than those produced by one of the most used techniques in the literature. We initially present an optimization cost function that proves to be better suited for the nonholonomic constraints of the vehicles that are the focus of this work. We also propose an improvement for the Alternating Algorithm, used to determinate the orientation angles on the calculation of the Dubins' Path through the waypoints. Finally, we use a recently developed optimization meta-heuristic, called Continuous-GRASP (C-GRASP), to generate a path that is shorter than paths obtained with classical techniques. Our results show significant improvements on the search for optimal paths for the case of nonholonomic vehicles. Our methodology was thoroughly evaluated and validated in simulation, and the results have shown a decrease on path's length of 28% on average, compared with the classic technique found in literature. In some cases a reduction of approximately 50% was obtained.**

I. Introduction

One of the major issues associated with path generation is strongly related to the mobility of the vehicle. A typical automobile, for example, cannot instantaneously move orthogonally to the wheel plane. Formally, this type of restriction on the combined rolling and sliding restrictions is known as nonholonomic constraints [1]. Indeed, a vast majority of actual vehicles are nonholonomic.

For these vehicles, there may exist one or more constraints associated with their movement. For instance, an airplane has as restriction its admissible curvature, maximum climb angle, non-zero translational velocity, among others. Therefore, path generation for nonholonomic vehicles requires specific approaches.

Nevertheless, the most fundamental requirement for a path is termed *feasibility*, that is, the planned path must be able to be executed by the proposed vehicle. During navigation, for example, we have to ensure that the vehicle's movement restrictions (i.e., nonholonomic constraints) will be correctly taken into account.

Another relevant issue is energy. The path must also be as short and smooth as possible in order to increase the vehicle's endurance. Therefore, several optimization issues lurk inside path planning problems which need to be adequately tackled so to enable mobile robots to efficient traverse the environment and fulfill their mission.

In this paper we present a set of strategies to reduce the length of paths for nonholonomic robots modeled as a Dubins' vehicle. We lay hold of a Traveling Salesman Problem (TSP) formulation in order to find the best sequence of waypoints in a two-dimensional space. We also provide an algorithm to calculate the best Dubins' circuit through these waypoints, and then use a meta-heuristic to improve the final results.

Section II presents previous work on the path generation problem. Next, on Section III, we show a cost function for an optimization problem that provides good results when applied planning paths for nonholonomic vehicles. We also present an optimization algorithm, based on a novel technique to set the angle between incoming and outgoing path segments at the points of TSP circuit (the circuit is precomputed). We then apply a meta-heuristic to improve the final results, by further reducing the total path length. A thorough comparison among all steps of our methodology is described on Section IV, so results are quantitatively analyzed.

II. Related Work

The generation of paths for nonholonomic vehicles is a topic of great importance, especially in robotics [1], [2]. The literature reports on several methods that address this, which is one of the central problems in robotics. However, a large number of those works has, as main focus, the generation of feasible paths (e.g. a path that satisfies the mobility constraints of a vehicle), but with no concern with the path's length.

Dubins, in his classical work [3], presented a technique to find the shortest path in the two-dimensional space, with orientation and curvature motion constraints between two adjacent points. The generated path is composed of straight line segments and arcs with radius that respect the curvature of the vehicle.

In [4] the authors propose a technique based on the use of Rapidly-Exploring Random Tree (RRT). The method allows to generate feasible paths for UAVs in environments with obstacles. At [5] a path planning methodology is proposed which is based on the use of genetic algorithms and Bézier curves.

If the length of the path becomes an issue, then minimization techniques should be readily invoked. Actually, the shortest path problem has been thoroughly studied in its most familiar instantiation: the TSP [6]. TSP is has been thoroughly investigated in the optimization literature. It can

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be simply described as: given a set $\mathcal{P}$ of waypoints (cities), starting from any given point, visit all the points in $\mathcal{P}$ exactly once and return to the starting point. Formally speaking, it is the problem of determinate the shortest Hamiltonian circuit on a set of points, which is a NP-Hard problem.

Therefore, the optimization issue in path planning for mobile robots is commonly mapped as a TSP instance. This is the case when considering a robot without nonholonomic constrains. The classical model of the TSP uses an objective function that minimizes the length of the total path through the waypoints reachable by the robot. The Euclidean distance is the choice objective function in most cases.

However, when dealing with nonholonomic robots, like Ackerman's car or fixed-wing UAVs, it is necessary to use a different metric to estimate the cost of the total circuit. Because of curvature constraints, paths for this type of vehicle are are generally composed of Dubins' curves [7], instead of straight lines.

Assuming, for instance, a robot with large inertia, rotations tend to be more expensive than translations, and this motivated [8] to propose a new model called Angular-metric TSP (TSPA). In their approach, besides minimizing the path's length, the total angular cost is also reduced. This cost is obtained by the summation of the angular differences between incoming and outgoing path segments at the waypoints.

Another issue with Dubins' curves is defining the orientations of the robot as it traverses the waypoints. [9] presents an algorithm called Alternating Algorithm that sets the orientations according to an heuristic that provides short (but not optimal) paths. [10] presents a probabilistic approximations in order to provide paths up to 15% shorter, but with less than 30 waypoints. As we will show in the next sessions, our method produces results that generate paths up to 28% shorter in most of the cases.

III. Methodology

Our methodology is composed of three sequential steps. First, we propose a cost function to the TSP optimization problem that, generally, provides better results when applied to path planning for nonholonomic vehicles with limited turning radius. Then we present an optimization algorithm based on a widely used technique to establish the orientation points to the TSP circuit previously calculated. Finally, we apply a meta-heuristic to improve the results obtained in the previous step in order to reduce the total path length.

A. Theoretical formalization

As far as the underlying physics of the system is concerned, the curvature may be defined as a quantity that is directly proportional to the lateral acceleration of the robot in the space. The value of κ_{max} is inversely proportional to the minimum curvature radius (ρ_{min}) of the curve that the vehicle is capable of execute. This is also proportional to the maximum lateral acceleration of the vehicle.

We also impose a velocity constraint to our vehicle, basically to avoid maneuvers that violate the Dubins' method between two adjacent waypoints.

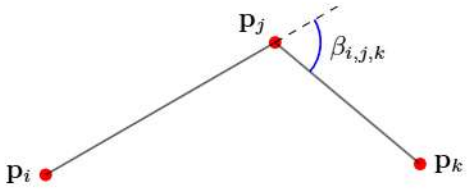


Fig. 1. Angular deviation from waypoints $\mathbf{p}_i$ to $\mathbf{p}_k$ through $\mathbf{p}_j$.

In order to be considered a feasible path, the curve $\vec{r}(t)$, between any two adjacent waypoints, must simultaneously fulfill curvature and velocity constraints of the vehicle:

$$\vec{r}(t) \rightarrow \quad |\kappa(t)| < \kappa_{max} \;\text{ and }\; \dot{\vec{r}}(t) \geq 0,$$

where $\kappa(t)$ is the curvature function of a curve in the n-dimensional space, that can be calculated by Equation 1.

$$\kappa(t) = \frac{|\dot{\vec{r}}(t) \times \ddot{\vec{r}}(t)|}{|\dot{\vec{r}}(t)|^3}. \tag{1}$$

B. Angular-metric TSP

In this section, we present an alternative optimization metric to be used in the TSP algorithm in lieu of the traditional Euclidean distance metric. The inspiration for our metric stems from the fact that the length of the Dubins' path that connects a pair of oriented waypoints tends to increase if the waypoints are not aligned, i.e., if the direction of the movement (waypoint orientation) differs significantly from one waypoint to the next. Therefore, our metric is based on the technique proposed in [11], which seeks to minimize changes in direction of connected waypoints.

For any three waypoints i, j and k, whose coordinates are $\mathbf{p}_i$, $\mathbf{p}_j$ and $\mathbf{p}_k$ respectively, we define the *angular deviation from i to k through j*, called $\beta_{i,j,k}$, as the angle between lines $\overline{\mathbf{p}_i\mathbf{p}_j}$ and $\overline{\mathbf{p}_j\mathbf{p}_k}$ (Fig. 1). Formally:

$$\beta_{i,j,k} \triangleq \arccos\left[\text{dir}(\mathbf{p}_i, \mathbf{p}_j) \cdot \text{dir}(\mathbf{p}_j, \mathbf{p}_k)\right], \tag{2}$$

where $\text{dir}(\mathbf{p}_m, \mathbf{p}_n)$ is defined as the direction (unit vector) from a given waypoint towards another, or:

$$\text{dir}(\mathbf{p}_m, \mathbf{p}_n) \triangleq \frac{\mathbf{p}_n - \mathbf{p}_m}{\|\mathbf{p}_n - \mathbf{p}_m\|}. \tag{3}$$

The final cost function is the sum of the individual angular deviation costs and the traditional Euclidean TSP cost. The former is scaled by the minimum turning radius of the robot, ρ_{min}: in this way, deviation costs are relatively more relevant for larger turning radii. Then, given the (unordered) set of waypoints $\mathcal{P}$ as input, the goal of the optimization process is to find a sequence of waypoints $\mathcal{C} = \langle \mathbf{p}_1, \ldots, \mathbf{p}_N \rangle$ that minimizes the total cost function:[1]

$$\mathcal{C} = \underset{\langle \mathbf{p}_1,\ldots,\mathbf{p}_N\rangle \in \text{permutations}(\mathcal{P})}{\arg\min} \Big[\rho_{\text{min}} \sum_i |\beta_{i-1,i,i+1}| + \\ + \sum_i \|\mathbf{p}_{i+1} - \mathbf{p}_i\|\Big]. \tag{4}$$

[1]For the sake of notational simplicity, since the sequence of waypoints forms a cyclic group we assume that $\mathbf{p}_m \equiv \mathbf{p}_n$ if $m \equiv n \pmod N$.

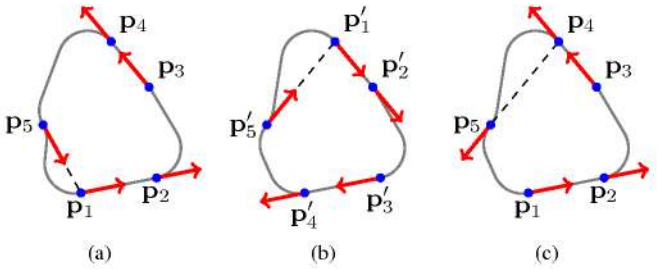


Fig. 2. Dubins' path obtained from Alternating Algorithm solutions. (a) Original circuit: the last ("dangling") waypoint ($\mathbf{p}_5$) points toward the first one ($\mathbf{p}_1$); (b) Reversed circuit, shifted in order to maintain the same dangling waypoint ($\mathbf{p}'_5$): While the straight segments and intermediate curves are exactly as the original, the change in orientation of the dangling waypoint completely changes the shape around it; (c) The reversed solution is equivalent to the original solution, however with $\mathbf{p}_5$ aligned with the previous waypoint and pointing away from it.

Notice that the first term is the cost due to the angular deviation and the second is the Euclidean TSP cost.

We call $\mathcal{C}$ the Dubins' circuit obtained by minimizing this objective function. Although this does not represent the optimum set of Dubins' paths through the waypoints, we assume it as a good approximation. It is necessary to say that this proposition is a simple heuristic based on the TSPA model proposed in [11]. We will demonstrate in the next section that, statistically, this model provides shorter Dubins' paths in average than those obtained using the Euclidean distance.

C. Best Alternating Algorithm (BAA)

Once the best sequence of waypoints $\mathcal{C}$ is defined, the next step is to determinate the orientation angles at each waypoint in order to calculate the corresponding Dubins' path.

One of the techniques most currently used in the literature, called the *Alternating Algorithm* [9], is actually a very simple heuristic. In short, it builds the circuit in such a way that consecutive pairs of waypoints are linked together with straight line segments, and the remaining (non-straight line) links are built using Dubins' curves based on the orientations previously assigned to the waypoints. Formally, the orientation of a waypoint n, expressed as ψ_n, is determined as follows:

$$\psi_n = \begin{cases} \text{dir}(\mathbf{p}_n, \mathbf{p}_{n+1}) & \text{if } n \text{ is odd} \\ \text{dir}(\mathbf{p}_{n-1}, \mathbf{p}_n) & \text{if } n \text{ is even,} \end{cases} \quad 1 \leq n \leq N. \quad (5)$$

However, the resulting circuit (hence its total length) depends on the arbitrary selection of the first waypoint, although, for example, $\mathcal{C} = \langle \mathbf{p}_1, \ldots, \mathbf{p}_N \rangle$ and $\mathcal{C} = \langle \mathbf{p}_2, \ldots, \mathbf{p}_N, \mathbf{p}_1 \rangle$ represent exactly the same cyclic circuit. Also, if the number of waypoints N is odd, then reversing the circuit (i.e., $\mathcal{C} = \langle \mathbf{p}_N, \mathbf{p}_{N-1}, \ldots, \mathbf{p}_1 \rangle$) results in another distinct solution. These observations and the paper [11] (which first present this idea) are the inspiration for our optimization technique, which we call, for short, *Best Alternating Algorithm* (BAA), detailed below:

- *If N is even:* In this case, there are only two different circuits: (i) when straight lines start at odd-numbered waypoints (the original proposition stated in Eq. (5)) and (ii) when straight lines start at even-numbered waypoints;
- *If N is odd:* In this case, there is a single waypoint that is not an end of any of the straight lines (as they link pairs of adjacent waypoints). Since it can be any one of the N waypoints, there are N possible solutions. Also, according to Eq. (5) this "dangling" waypoint is oriented toward the following neighbor; then, if one analyzes the circuit with waypoints in reversed order (Fig. 2), the dangling waypoint will be oriented away from the previous neighbor, which accounts for another N possible solutions, giving a grand total of $2N$ solutions.

In any case, all solutions must be analyzed and the shortest one is picked as the solution of our algorithm. The orientation of a waypoint i will be referred henceforth as ψ_i and the whole sequence of orientations (of the best solution) as $\Psi = \langle \psi_1, \ldots, \psi_N \rangle$.

D. Continuous-GRASP

As presented in the previous section, the Alternating Algorithm, as well as our Best Alternating Algorithm, are used to assign orientation to the points. However, the assigned orientation are frequently far from the value that would make it possible to find the shortest path.

Thus, the main objective of this section is to describe the technique used to optimize the orientation angles obtained from the BAA, and by that reducing the overall length of the path found. To accomplish this task we will use the C-GRASP algorithm.

The meta-heuristic Greedy Randomized Adaptive Search Procedures (GRASP) is described in [12] as a multi-start local search procedure, where each GRASP iteration consists of two phases, a construction phase and a local search phase.

The C-GRASP method was proposed in [13], which extended the GRASP from the domain of discrete optimization to that of continuous global optimization. It is often a good alternative to deal with global optimization problems, and it is also easy to describe and implement, since it doe not depend on derivative nor on *a priori* information. The C-GRASP heuristic almost always converges to a good solution in a very short amount of time.

C-GRASP works by discretizing the domain into a uniform grid, where both the construction and local improvement phases move along points on the grid. As the algorithm is executed, the grid adaptively becomes denser.

We used the implementation of the C-GRASP presented in [14], which aims at speeding up the execution and increasing robustness. The pseudo-code for C-GRASP adapted to the context being treated here is presented in Algorithm 1.

The objective function $f(\cdot)$ is equal to the length of the new Dubins' path, based on the new orientation angles being calculated (x). Since we already have a good estimation for the path, instead of initializing f^* (best solution found) to infinity (as proposed in the classic implementation of the C-

Algorithm 1 C-GRASP($\mathcal{C}, h_s, h_e, p_{lo}$)

$\Psi \leftarrow$ BAA($\mathcal{C}$)
$f^* \leftarrow f(\mathcal{C}, \Psi)$
while Stopping criteria not met **do**
  $x \leftarrow \Psi$
  $h \leftarrow h_s$
  while $h > h_e$ **do**
    $Imp_C \leftarrow 0, Imp_L \leftarrow 0$
    $(x, Imp_C) \leftarrow$ constructionPhase(x, h)
    $(x, Imp_L) \leftarrow$ localSearchPhase(x, h, p_{lo})
    if $f(\mathcal{C}, x) < f^*$ **then**
      $x^* \leftarrow x, f^* \leftarrow f(\mathcal{C}, x)$
    end if
    if $Imp_C = 0$ **and** $Imp_L = 0$ **then**
      $h \leftarrow h/2$
    end if
  end while
end while
return x^*

GRASP technique), we initialize it with the length of the path found by the TSPA and the BAA.

The parameters h_s and h_e define the starting and ending grid discretization (in our case the angle step), while parameter p_{lo} defines the portion of the neighborhood of the current solution that is searched during the local improvement phase.

The initial solution x is set to the Ψ values calculated by the BAA. This values represent a good initial estimate for the optimization phase.

The construction phase generates an acceptable solution by randomly selecting the elements over a list that is ordered by an adaptive greed function. In the local search phase, a local search is executed in the neighborhood of given calculated solution. While an improvement of the solution is occurring, the local search procedure continues until a local optimal is found.

Variables Imp_C and Imp_L verify if an improvement has occurred in the construction phase or during local search. If no improvement was perceived, the level of discretization used is divided by two. The variable Imp_C will be false only if there has been no improvement in the solution in the construction phase. The variable Imp_L will be false only if the input used in the local search phase is already a local minimum (h-local minimum).

Due to space restrictions, we refer the reader to the material found in [13], [14] for further details on the construction and local improvement phase.

IV. Experiments

The experiments were performed with 15 nodes randomly placed in an environment without obstacles. We also chose a simulated vehicle with minimum curvature radius $\rho_{\min} = 50\,\mathrm{m}$, which is significant when compared to the environment dimensions.

The following values were assigned to the C-GRASP (Algorithm 1) constants: $h_s = \pi\,\mathrm{rad}$, $h_e = 0.01\,\mathrm{rad}$ and $p_{lo} = 0.7$. These values were empirically chosen and provided good results with low convergence time.

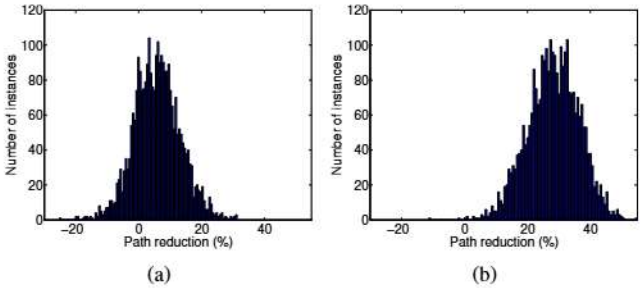


Fig. 3. Histograms of the relative path's length reduction using different methods: (a) Improvement of TSPA+BAA over TSP+BAA; (b) Improvement of TSPA+BAA+C-GRASP over TSP+Alternating Algorithm.

As the proposed technique is statistical in nature, a large number of experiments is necessary to validate the improvements. Thus, we have run a total of 5000 experiments.

A large percentage of the circuits resulting from the TSP and from the TSPA optimization process are the same. For the results depicted in Figure 3, only the instances whose TSP and TSPA are different (about 60% of the test cases) were used.

Figure 3(a) presents the histogram of the reduction of the Dubins' path length when the path is produced by the proposed BAA over the TSP circuit and the TSPA circuit. In about 80% of the cases, results obtained with the BAA from a TSPA circuit are better than those obtained from a TSP circuit. Since the waypoints are better aligned using the TSPA method, the generated Dubins' path also tends to be shorter. However, sometimes the benefits of the reduced angular cost (using the TSPA method) can not compensate for the increase of the path's length, which explains the worse results of the remaining 20% of the cases.

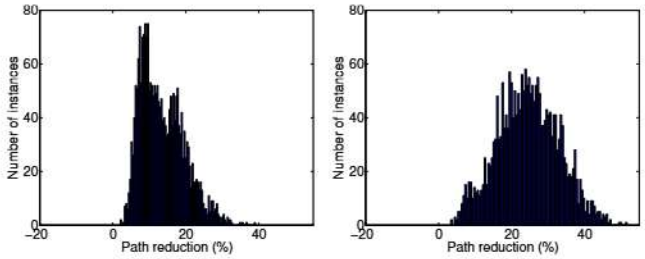
Figure 3(b) shows the histogram of the reduction of path length using the classic TSP for Dubins' Vehicle (TSPD) (TSP with Alternating Algorithm) and the result of our entire methodology (TSPA + BAA + C-GRASP). The latter improvement has a mean value of 28.2% and a standard deviation of 8.1%.

Figure 4 presents the analysis of the percentage of experiments when the circuit obtained by the TSPA is equal to the TSP circuit (i.e., when the angular cost function produced the same circuit as the original TSP).

It is important to notice that even in the cases where there is no gain in angular cost of the TSPA when compared with TSP, the subsequent steps (BAA and C-GRASP) are able to improve the results.

In all experiments involving both Euclidean and angular metrics, we have used the IBM ILOG CPLEX® version 12.2 solver to compute the optimum circuit through all waypoints.

Next we describe a sample case demonstrating our method. Figure 5(a) presents the initial circuit obtained using the classic TSP with Euclidean distance, which refers to the globally optimal solution (not considering the curvature constraint). Figure 5(b) shows another circuit, produced by



(a) (b)

Fig. 4. Histograms of the relative path's length reduction for equal TSP and TSPA circuits: (a) Improvement of TSPA+BAA+C-GRASP over TSP+BAA; (b) Improvement of TSPA+BAA+C-GRASP over TSP+Alternating Algorithm.

our methodology with the metric of Equation 4.

Figure 5(c) presents the Dubins' path for the waypoint sequence of Figure 5(a), using the classic Alternating Algorithm. The corresponding result is presented in Figure 5(d) based on the sequence of 5(b).

The results obtained with our BAA methodology are shown in figures 5(e) and 5(f). The first one is based on the sequence of Figure 5(a), while the other is based on Figure 5(b).

Although the circuit of Figure 5(b) is longer than the one shown in Figure 5(a), the waypoints are better aligned (which can be observed by the decrease of the angular cost), which promotes Dubins' segments with less curves – an effect that can be observed in both Figures 5(d) and 5(f) when compared respectively with Figures 5(c) and 5(e). Also, notice that the BAA method is able to produce shorter paths when compared to those obtained with the classic Alternating Algorithm.

Finally, Figures 5(g) and 5(h) present the final Dubins' path obtained by the angular optimization presented in Section III-D for both the TSP and the TSPA cases, respectively. Results show that the C-GRASP technique can eliminate unnecessary loops within the circuit, thereby significantly reducing the total path's length. Also, notice that the angular metric cost function proposed in Section III-B leads to a final result that has much less curves.

V. Conclusion and Future Work

This work presented a set of strategies to reduce the path's length for Dubins' vehicles. The methodology presented good results, reducing the path's length of 28.2% on average when compared with classic techniques in the literature. In some cases, we obtained a reduction of up to 50%.

Also, the circuit produced by the Angular-metric TSP has less turns at the waypoints, which significantly decreases the length of the Dubins' curves. The use of Euclidean distance as cost function in the optimization problem not necessarily provides good results, since the path length for nonholonomic vehicles depends on the orientation of the vehicle at the waypoints.

The work has several directions that will be addressed next to improve the proposed methodology. Initially, we intend to extend the technique to environments containing obstacles.

Small variations in the orientation of the vehicle can have great impact on the path's length. Thus, we intend to investigate improvements for the Alternating Algorithm, so that good initial estimates of orientation can be obtained.

One of the main problems concerning the use of Dubins' curves is the discontinuity on the curvature profile. This actually leads to abrupt lateral accelerations leading to paths that are not feasible for a real robot. This has lead us to investigation on the use of another path planning technique in the context of this optimization problem.

VI. Acknowledgments

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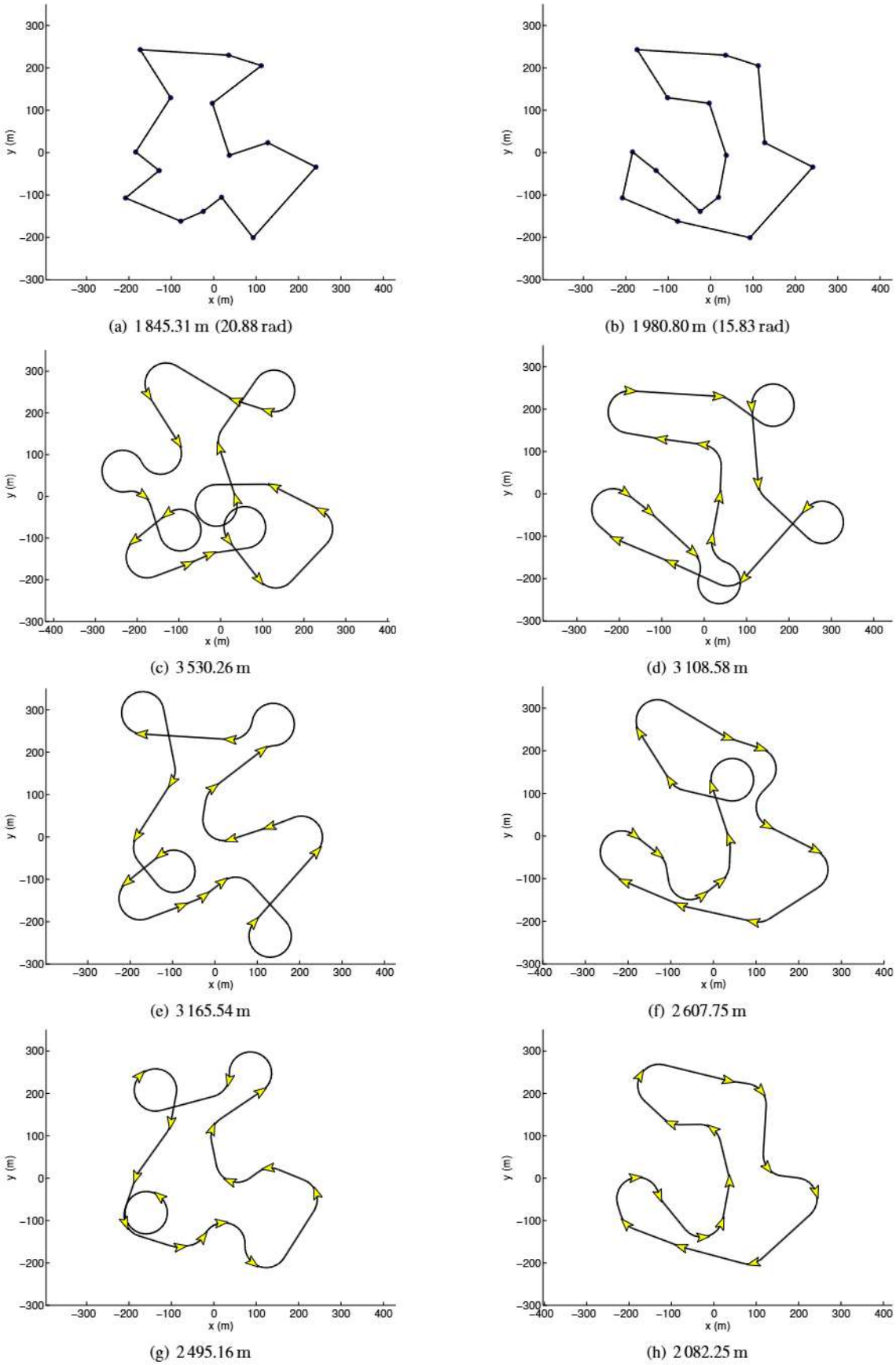
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(a) 1 845.31 m (20.88 rad)

(b) 1 980.80 m (15.83 rad)

(c) 3 530.26 m

(d) 3 108.58 m

(e) 3 165.54 m

(f) 2 607.75 m

(g) 2 495.16 m

(h) 2 082.25 m

Fig. 5. One example case of our methodology: (a) circuit using TSP metrics, (b) circuit using TSPA metrics, (c) Alternating Algorithm applied to the TSP circuit, (d) Alternating Algorithm applied to the TSPA circuit, (e) BAA algorithm applied to the TSP circuit, (f) BAA algorithm applied to the TSPA circuit, (g) TSP and BAA algorithm with C-GRASP and (h) TSPA and BAA algorithm with C-GRASP. The value presented for each graphic is the total path length, and the angle within parentheses is the total angular cost.